

Odd- κ Symmetry of Finite-Order Threshold Sets

Candidate full answer to an open problem in arXiv:2201.09545

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Status. Candidate full solution, likely valid, pending human review. For every dimension $d \geq 1$, every $\kappa \geq 1$, and every $m \geq 0$,

$$\Theta_{m,\kappa}(\Delta) = -\Theta_{m,\kappa}(\Delta).$$

This supplies the odd- κ case left open after Lemma 3.3 of the source.

1 Source question

Golénia and Mandich prove the displayed symmetry for even κ and ask whether it also holds for odd κ [1, p. 12, after Lemma 3.3].

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The proof of Lemma 3.3 follows directly from the definition of $\Theta_{m,\kappa}(\Delta)$. It is an open problem for us to decide if Lemma 3.3 also holds for κ odd. We also take the opportunity to prove Lemma 1.17.

Proof of Lemma 1.17. Let $E \in \Theta_{m,\kappa}(\Delta[d])$, with $E = x_{q,1} + \dots + x_{q,d}$ for $0 \leq q \leq m$. Set $\mathcal{E} = E + \cos(l\pi/\kappa)$, $0 \leq l \leq \kappa$. Then $\forall j \in \mathbb{N}^*$ (and using the notation (1.8) instead of (3.3)) :

$$g_{j\kappa}(\vec{x}_m, \cos(l\pi/\kappa)) = g_{j\kappa}(\vec{x}_m) = \sum_{q=0}^{m-1} \omega_q \cdot g_{j\kappa}(\vec{x}_q) = \sum_{q=0}^{m-1} \omega_q \cdot g_{j\kappa}(\vec{x}_q, \cos(l\pi/\kappa)).$$

This implies $\mathcal{E} \in \Theta_{m,\kappa}(\Delta[d+1])$. □

Figure 1: The open problem in the source, arXiv PDF page 12.

For $\vec{x} = (x_1, \dots, x_d) \in [-1, 1]^d$, the source defines

$$g_{j\kappa}(\vec{x}) = \sum_{i=1}^d (1 - x_i^2) U_{j\kappa-1}(x_i), \quad j \in \mathbb{N}^*, \quad (1)$$

where U_n is the Chebyshev polynomial of the second kind. The constant energy surface is

$$S_E = \{\vec{x} \in [-1, 1]^d : x_1 + \dots + x_d = E\}.$$

For $m \geq 1$, membership $E \in \Theta_{m,\kappa}(\Delta)$ means that there are $\vec{x}_0, \dots, \vec{x}_m \in S_E$ and real weights $\omega_0, \dots, \omega_{m-1} \leq 0$ such that

$$g_{j\kappa}(\vec{x}_m) = \sum_{q=0}^{m-1} \omega_q g_{j\kappa}(\vec{x}_q) \quad \text{for every } j \in \mathbb{N}^*. \quad (2)$$

For $m = 0$, the definition asks for $\vec{x}_0 \in S_E$ with $g_{j\kappa}(\vec{x}_0) = 0$ for every j .

2 Proof intuition

Negate every coordinate of every witness point. This changes the energy from E to $-E$. The parity of $U_{j\kappa-1}$ gives a scalar sign which may depend on j when κ is odd, but for each fixed j it is identical at every witness point. Consequently it multiplies both sides of (2) by the same number and cannot disturb the witness relation or the signs of its weights.

3 The theorem and proof

Theorem 1. *For all $d, \kappa \in \mathbb{N}^*$ and $m \in \mathbb{N}$,*

$$\Theta_{m,\kappa}(\Delta) = -\Theta_{m,\kappa}(\Delta).$$

In particular, Lemma 3.3 of [1] holds when κ is odd.

Proof. The standard parity identity for Chebyshev polynomials is

$$U_n(-t) = (-1)^n U_n(t).$$

Because $1 - t^2$ is even, (1) yields

$$g_{j\kappa}(-\vec{x}) = (-1)^{j\kappa-1} g_{j\kappa}(\vec{x}) \quad (j \in \mathbb{N}^*). \quad (3)$$

Suppose first that $m \geq 1$ and $E \in \Theta_{m,\kappa}(\Delta)$, with witnesses $\vec{x}_0, \dots, \vec{x}_m$ and weights $\omega_q \leq 0$ as in (2). Put $\vec{y}_q = -\vec{x}_q$. Then $\vec{y}_q \in S_{-E}$. For every fixed j , multiply (2) by $(-1)^{j\kappa-1}$ and use (3); this gives

$$g_{j\kappa}(\vec{y}_m) = \sum_{q=0}^{m-1} \omega_q g_{j\kappa}(\vec{y}_q).$$

The same weights are still nonpositive, so they witness $-E \in \Theta_{m,\kappa}(\Delta)$.

If $m = 0$, choose $\vec{x}_0 \in S_E$ with $g_{j\kappa}(\vec{x}_0) = 0$ for all j . Equation (3) shows that $-\vec{x}_0 \in S_{-E}$ has the same zero property. Thus in every case $\Theta_{m,\kappa}(\Delta) \subseteq -\Theta_{m,\kappa}(\Delta)$. Applying the same argument to $-E$ gives the reverse inclusion. $\square$

Remark 1. *For odd κ , the factor in (3) is $(-1)^{j-1}$ and hence depends on j . This causes no difficulty: the definition requires a separate equality for each j , and within that equality the factor is common to every witness point. This is precisely the point that removes the parity restriction in the source lemma.*

4 Verification, scope, and novelty boundary

The companion script `code/verify_parity.py` constructs the U_n by their recurrence and symbolically verifies (3) for 224 triples of dimension, κ , and j . This is only a transcription diagnostic; the proof above is exact and finite.

The theorem answers only the source's odd- κ symmetry question for the finite-order sets $\Theta_{m,\kappa}(\Delta)$. It does not identify the full threshold set, prove its countability, or establish the conjectured Mourre bands.

Bounded exact-sentence, arXiv-id, author, lemma-label, and keyword searches through 26 August 2026 found the source and companion papers, but no later primary source recording this answer or parity proof. Priority is not certified and expert review remains appropriate.

5 Human review recommendation

Check the source definition (2) and confirm that it is imposed separately for each j . Once this is verified, simultaneous negation and Chebyshev parity prove the result without any analytic or convergence input.

References

- [1] S. Golénia and M.-A. Mandich, *Thresholds and more bands of a.c. Spectrum for the discrete Schrödinger operator with a more general long range condition*, arXiv:2201.09545 (2022).